

Multi-Pod Security Center

Automated Threat Intelligence for Distributed Nodes

A lightweight, low-latency framework for real-time monitoring, anomaly detection, and centralized data visualization.

THE CHALLENGE

Monitoring Gap in Pod Architectures

| Visibility Issues

Distributed systems suffer from "monitoring blindness" where localized file system anomalies or unauthorized storage growth go undetected until critical failure occurs.

| Resource Constraints

Standard agents (e.g., Datadog, Prometheus) can be resource-heavy. Developers need a non-intrusive solution that operates with **minimal CPU/Memory overhead**.

Decoupled Full-Stack Ecosystem

Backend Infrastructure

- **FastAPI:** High-performance asynchronous API for telemetry ingestion.
- **SQLite:** Transactional storage with optimized indexing on timestamps and alert levels.
- **Pydantic:** Strict schema validation for incoming node reports.

Command Interface

- **Streamlit:** Modern interactive dashboard for SOC operators.
- **Plotly:** Dynamic vector charts for distribution analysis.
- **Live Feed:** Real-time filtering and status mapping for rapid response.

Deterministic Anomaly Classification

The system utilizes a **Threshold-Based Engine** to categorize reports in real-time. This eliminates manual audit fatigue by instantly flagging critical growth.

```
# core/logic.py
THRESHOLD = os.getenv("ALERT_THRESHOLD_GB", 0.01)

def classify(items):
    is_critical = any(i.size_gb > THRESHOLD for i in items)
    return "CRITICAL" if is_critical else "NORMAL"
```

Optimized Relational Storage Schema

To handle high-frequency reporting from multiple pods, the database layer implements specific optimizations:

- **Atomic Operations:** Ensuring report integrity across concurrent uploads.
- **Foreign Key Constraints:** Linking reports to detailed sub-folder items.
- **Index Optimization:** PRAGMA settings for maximum read/write performance.

O(1)
INGESTION

Async
I/O MODEL

Fast
FILTERING

Rel.
STABILITY

Real-time Operational Visibility

| Live Monitoring

The "Live Feed" tab provides an auto-refreshing stream of events. Critical status is **color-mapped to high-contrast yellow** for immediate visual capture.

| Filtering & Search

Supports granular multi-select filters for Hostnames and Risk Levels, allowing operators to isolate threats in complex environments within seconds.

Quantitative Security Metrics

Beyond raw logs, the system synthesizes data into actionable intelligence through Plotly integration.

- **Risk Distribution:** Ratio of Normal vs. Critical reports.
- **Node Performance Bar:** Comparative analysis of storage utilization.
- **Historical Trends:** Tracking anomalous growth patterns over time.

KEY DIFFERENTIATORS

Performance & Scalability Benchmarks

| Zero Configuration

Native Python implementation means no external agents or heavy sidecars are required. Just a simple HTTP POST to report data.

| Asynchronous Efficiency

FastAPI's event loop allows the system to handle thousands of concurrent pod connections without blocking, ensuring high scalability.

Transcendent Security Evolution

1. **AI Integration:** Predictive modeling to forecast storage exhaustion before it happens.
2. **Security Hardening:** JWT-based authentication for agent-to-backend communication.
3. **Multi-Tenant Support:** Logical separation of data for enterprise-level deployments.
4. **Auto-Remediation:** Triggering cleanup scripts via API webhooks.